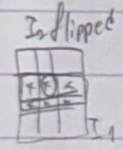
ETA 445 - Εργασία 4 - Μέρος Α

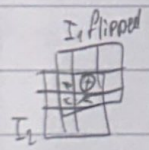
$$I_1 = \begin{bmatrix} 7 & 4 & 8 \\ 15 & 0 & 2 \\ 6 & 12 & 1 \end{bmatrix} \qquad I_2 = \begin{bmatrix} 0{,}0251 & 0{,}1453 & 0{,}0251 \\ 0{,}1453 & 0{,}3183 & 0{,}1453 \\ 0{,}0251 & 0{,}1453 & 0{,}0251 \end{bmatrix}$$

$$J(i,j) = I_1(i,j) * I_2(i,j) = \sum_{m=0}^{2} \sum_{n=0}^{2} I_1(m,n) \cdot I_2(i-m, j-m)$$

$$J(0,0) = I_1(0,0) \cdot I_2(0,0) + I_1(0,1) \cdot I_2(0,-1) + \cancel{I_1(0,2) \cdot I_2(0,-2)}^{0} +$$
$$I_1(1,0) \cdot I_2(-1,0) + I_1(1,1) \cdot I_2(-1,-1) + \cancel{I_1(1,2) \cdot I_2(-1,-2)}^{0} +$$
$$\cancel{I_1(2,0) \cdot I_2(-2,0)}^{0} + I_1(2,1) \cdot I_2(-2,-1) + \cancel{I_1(2,2) \cdot I_2(-2,-2)}^{0}$$
$$= 7 \cdot 0{,}3183 + 4 \cdot 0{,}1453 + 15 \cdot 0{,}1453 + 0 \cdot 0{,}0251 = \boxed{4{,}9869}$$



$$J(0,1) = I_1(0,0) \cdot I_2(0,1) + I_1(0,1) \cdot I_2(0,0) + I_1(0,2) \cdot I_2(0,-1) +$$
$$I_1(1,0) \cdot I_2(-1,1) + I_1(1,1) \cdot I_2(-1,0) + I_1(1,2) \cdot I_2(-1,-1) +$$
$$\cancel{I_1(2,0) \cdot I_2(-2,1)}^{0} + \cancel{I_1(2,1) \cdot I_2(-2,0)}^{0} + \cancel{I_1(2,2) \cdot I_2(-2,-1)}^{0}$$
$$= 7 \cdot 0{,}1453 + 4 \cdot 0{,}3183 + 8 \cdot 0{,}1453 + 15 \cdot 0{,}0251 + 0 \cdot 0{,}1453 + 2 \cdot 0{,}0251 = \boxed{3{,}8779}$$



$$J(0,2) = \cancel{I_1(0,0) \cdot I_2(0,2)}^{0} + I_1(0,1) \cdot I_2(0,1) + I_1(0,2) \cdot I_2(0,0) +$$
$$\cancel{I_1(1,0) \cdot I_2(-1,2)}^{0} + I_1(1,1) \cdot I_2(-1,1) + I_1(1,2) \cdot I_2(-1,0) +$$
$$\cancel{I_1(2,0) \cdot I_2(-2,2)}^{0} + \cancel{I_1(2,1) \cdot I_2(-2,1)}^{0} + \cancel{I_1(2,2) \cdot I_2(-2,0)}^{0}$$
$$= 4 \cdot 0{,}1453 + 8 \cdot 0{,}3183 + 0 \cdot 0{,}0251 + 2 \cdot 0{,}1453 = \boxed{3{,}4176}$$

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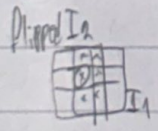

$$J(1,0) = I_1(0,0) \cdot I_2(1,0) + I_1(0,1) \cdot I_2(1,-1) + \cancel{I_1(0,2) \cdot I_2(1,-2)}^{0} +$$
$$I_1(1,0) \cdot I_2(0,0) + I_1(1,1) \cdot I_2(0,-1) + \cancel{I_1(1,2) \cdot I_2(0,-2)}^{0} +$$
$$I_1(2,0) \cdot I_2(-1,0) + I_1(2,1) \cdot I_2(-1,-1) + \cancel{I_1(2,2) \cdot I_2(-1,-2)}^{0}$$
$$= 7 \cdot 0{,}1453 + 4 \cdot 0{,}0251 + 15 \cdot 0{,}3183 + 0 \cdot 0{,}1453 + 6 \cdot 0{,}1453 + 12 \cdot 0{,}0251 = \boxed{7{,}0637}$$

$$J(1,1) = I_1(0,0) \cdot I_2(1,1) + I_1(0,1) \cdot I_2(1,0) + I_1(0,2) \cdot I_2(1,-1) +$$
$$I_1(1,0) \cdot I_2(0,1) + I_1(1,1) \cdot I_2(0,0) + I_1(1,2) \cdot I_2(0,-1) +$$
$$I_1(2,0) \cdot I_2(-1,1) + I_1(2,1) \cdot I_2(-1,0) + I_1(2,2) \cdot I_2(-1,-1) \qquad \boxed{= 5{,}3438}$$
$$= 7 \cdot 0{,}0251 + 4 \cdot 0{,}1453 + 8 \cdot 0{,}0251 + 15 \cdot 0{,}1453 + 0 \cdot 0{,}3183 + 2 \cdot 0{,}1453 + 6 \cdot 0{,}0251 + 12 \cdot 0{,}1453 + 1 \cdot 0{,}0251$$

$$J(1,2) = \cancel{I_1(0,0) \cdot I_2(1,2)}^{0} + I_1(0,1) \cdot I_2(1,1) + I_1(0,2) \cdot I_2(1,0) +$$
$$\cancel{I_1(1,0) \cdot I_2(0,2)}^{0} + I_1(1,1) \cdot I_2(0,1) + I_1(1,2) \cdot I_2(0,0) +$$
$$\cancel{I_1(2,0) \cdot I_2(-1,2)}^{0} + I_1(2,1) \cdot I_2(-1,1) + I_1(2,2) \cdot I_2(-1,0)$$
$$= 4 \cdot 0{,}0251 + 8 \cdot 0{,}1453 + 0 \cdot 0{,}1453 + 8 \cdot 0{,}3183 + 12 \cdot 0{,}0251 + 1 \cdot 0{,}1453 = \boxed{12{,}345}$$

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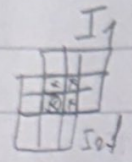
Zuigzag:

$$I_1 = \begin{bmatrix} 7 & 4 & 8 \\ 15 & 0 & 2 \\ 6 & 12 & 1 \end{bmatrix} \qquad I_2 = \begin{bmatrix} 0{,}0251 & 0{,}1453 & 0{,}0251 \\ 0{,}1453 & 0{,}3183 & 0{,}1453 \\ 0{,}0251 & 0{,}1453 & 0{,}0251 \end{bmatrix}$$



$$J(2,0) = \cancel{I_1(0,0)\cdot I_2(2,0)}^{0} + \cancel{I_1(0,1)\cdot I_2(2,-1)}^{0} + \cancel{I_1(0,2)\cdot I_2(2,-2)}^{0} +$$
$$I_1(1,0)\cdot I_2(1,0) + I_1(1,1)\cdot I_2(1,-1) + \cancel{I_1(1,2)\cdot I_2(1,-2)}^{0} +$$
$$I_1(2,0)\cdot I_2(0,0) + I_1(2,1)\cdot I_2(0,-1) + \cancel{I_1(2,2)\cdot I_2(0,-2)}^{0}$$
$$= 15\cdot 0{,}1453 + 0\cdot 0{,}0251 + 6\cdot 0{,}3183 + 12\cdot 0{,}1453 = \boxed{5{,}8302}$$

$$J(2,1) = \cancel{I_1(0,0)\cdot I_2(2,1)}^{0} + \cancel{I_1(0,1)\cdot I_2(2,0)}^{0} + \cancel{I_1(0,2)\cdot I_2(2,-1)}^{0} +$$
$$I_1(1,0)\cdot I_2(1,1) + I_1(1,1)\cdot I_2(1,0) + I_1(1,2)\cdot I_2(1,-1) +$$
$$I_1(2,0)\cdot I_2(0,1) + I_1(2,1)\cdot I_2(0,0) + I_1(2,2)\cdot I_2(0,-1)$$
$$= 15\cdot 0{,}0251 + 0\cdot 0{,}1453 + 2\cdot 0{,}0251 + 6\cdot 0{,}1453 + 12\cdot 0{,}3183 + 1\cdot 0{,}1453 = \boxed{5{,}2627}$$

$$J(2,2) = \cancel{I_1(0,0)\cdot I_2(2,2)}^{0} + \cancel{I_1(0,1)\cdot I_2(2,1)}^{0} + \cancel{I_1(0,2)\cdot I_2(2,0)}^{0}$$
$$\cancel{I_1(1,0)\cdot I_2(1,2)}^{0} + I_1(1,1)\cdot I_2(1,1) + I_1(1,2)\cdot I_2(1,0)$$
$$\cancel{I_1(2,0)\cdot I_2(0,2)}^{0} + I_1(2,1)\cdot I_2(0,1) + I_1(2,2)\cdot I_2(2,0)$$
$$= 0\cdot 0{,}0251 + 2\cdot 0{,}1453 + 12\cdot 0{,}1453 + 1\cdot 0{,}3183 = \boxed{2{,}3511}$$

$$\boxed{J = \begin{bmatrix} 4{,}9869 & 3{,}8179 & 3{,}4176 \\ 7{,}0631 & 5{,}3438 & 2{,}345 \\ 5{,}8302 & 5{,}2627 & 2{,}3511 \end{bmatrix}}$$

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